PRIVACY-ENHANCED SYNTHETIC DATA GENERATION FOR AUTONOMOUS VEHICLE TRAINING

INDIVIDUAL THESIS

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Individual component: Privacy Integration

Group ID: R26-CS-010

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science (Honours) Degree in Information Technology

Specialising in Cyber Security

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Perception models for autonomous vehicles are trained on large amounts of labeled road data, and this data records people. Simulators such as CARLA reduce the need for real footage. However, they are tuned with real statistics, and trained models can still reveal their training records. This thesis presents the Privacy Integration component of a four-part framework for privacy-enhanced synthetic training data. The component applies differential privacy where real information enters the pipeline. It then tests this protection with attacks. At calibration, the Laplace mechanism adds noise to seven driving statistics under a total budget of $\varepsilon = 1.8$. At training, Opacus DP-SGD clips gradients, adds Gaussian noise and stops at a hard budget limit. The component then attacks its own models. It records the results in HMAC-signed reports and a privacy certificate. The evaluation used 10,000 synthetic frames. A non-private baseline and three budgets ($\varepsilon = 1$, 3 and 8) were each trained with three seeds. The shadow-model attack reached an AUC of 0.639 against the baseline. Against the private models, it fell to between 0.504 and 0.515 ($p \leq 0.0032$). The cost was 0.16 to 0.21 in test AUC. The baseline leaked only moderately, so a stress test used a deliberately overfitted model and five seeds. In this test, the attack AUC fell from 0.628 to 0.501 at $\varepsilon = 1$ ($p = 0.00004$). This removed 99.5% of the attacker's advantage, and smaller budgets gave significantly stronger protection. The Risk and Compliance component verified all twelve signed reports and budget logs. Its composition check found an understated calibration budget, which was corrected. The results show that a formal ε should be tested with an attack. That attack should first succeed against a non-private model.

Keywords: differential privacy, DP-SGD, membership inference, synthetic data, autonomous vehicles

ACKNOWLEDGEMENTS

I would like to thank my supervisor, **Prof. Anuradha Jayakody**, and my co-supervisor, **Mr. Dinith Primal**, for their guidance and feedback throughout this research. Their early advice helped me narrow the privacy work to a scope I could build and test in time.

I also thank my team members, Yubitha S. (IT22541284), Apeksha M.K.S.R. (IT22110220) and Wijesinghe R.D.P.T. (IT22066916), for their support and cooperation. Another member checked my outputs independently, and this check shaped many design decisions in this thesis. I thank the members of the progress review panels. Their comment on the size of the privacy effect led to the stress test in this thesis. Finally, I thank the developers of PyTorch, Opacus and CARLA. Their open-source tools made this work possible.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AES	Advanced Encryption Standard
AI	Artificial Intelligence
AUC	Area Under the ROC Curve
AV	Autonomous Vehicle
CARLA	Car Learning to Act (open-source driving simulator)
CBC	Cipher Block Chaining
CNN	Convolutional Neural Network
DP	Differential Privacy
DP-SGD	Differentially Private Stochastic Gradient Descent
FID	Fréchet Inception Distance
FPR	False-Positive Rate
GCM	Galois/Counter Mode
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
HMAC	Hash-based Message Authentication Code
JSON	JavaScript Object Notation
KITTI	Karlsruhe Institute of Technology and Toyota Technological Institute (dataset)
mAP	Mean Average Precision
MIA	Membership Inference Attack
NIST	National Institute of Standards and Technology
PDF	Portable Document Format
QR	Quick Response (code)
RDP	Rényi Differential Privacy
ROC	Receiver Operating Characteristic
RQ	Research Question
SD	Standard Deviation
SDG	Sustainable Development Goal
SGD	Stochastic Gradient Descent
SHA	Secure Hash Algorithm
TPR	True-Positive Rate

Abbreviation	Meaning
YOLO	You Only Look Once (object detector)

1 INTRODUCTION

1.1 Overall project context

Autonomous vehicles use machine learning to detect objects, predict the movement of other road users and plan their driving. These models need large amounts of labeled sensor data. Public datasets such as KITTI [1] and the Waymo Open Dataset [2] were built by driving sensor-equipped cars on public roads. The recordings were then labeled by hand. This data contains personal information. Cameras record faces and license plates. Repeated routes can also show where people live and work. Under the General Data Protection Regulation (GDPR), recordings of identifiable people are personal data. Article 25 of the GDPR requires data protection by design and by default [3].

One response is to generate training data in a simulator such as CARLA [4]. A simulator removes the need to record real people. However, it does not remove privacy risk on its own. The simulator must be calibrated with real observations, such as traffic density, weather and time of day. This means real information still enters the pipeline. Models trained on the generated data can also reveal which records were in their training set [5], [6]. Stadler et al. [7] showed that synthetic data does not protect its source records without a formal privacy mechanism.

Project R26-CS-010 addresses these risks with a privacy-enhanced synthetic data generation framework. The framework has four components. Privacy Integration (IT22309556) applies and tests differential privacy where real information enters the pipeline. Dataset Generation (IT22541284) captures CARLA frames. It protects every file with SHA-256 hashing, AES-256 encryption and role-based access control. Utility Evaluation (IT22110220) checks the data for tampering and measures its value for training a detector. Risk and Compliance (IT22066916) verifies the privacy controls with its own checks and a reconstruction attack. It also maps the evidence to GDPR and NIST requirements and issues a signed compliance certificate.

1.2 Individual component context

This thesis covers the Privacy Integration component. Real information can enter the pipeline at two points, and this component is responsible for both. The first point is calibration, where real aggregate statistics configure the simulator. The second point is model training, where a model may memorize its training records. The component applies differential privacy at each point. It then tests whether the protection holds against an attacker.

The component sits between the data owner and the rest of the framework. It receives aggregate statistics on the data owner's machine. Only privatized values are sent to Dataset Generation. Later, the component receives encrypted frames from Dataset Generation and clean labels from Utility Evaluation. It trains models on this data, attacks the models and sends signed evidence to Risk and Compliance. Two other components use its outputs, and a third component checks its claims. Figure 1.1 shows how the four components connect.

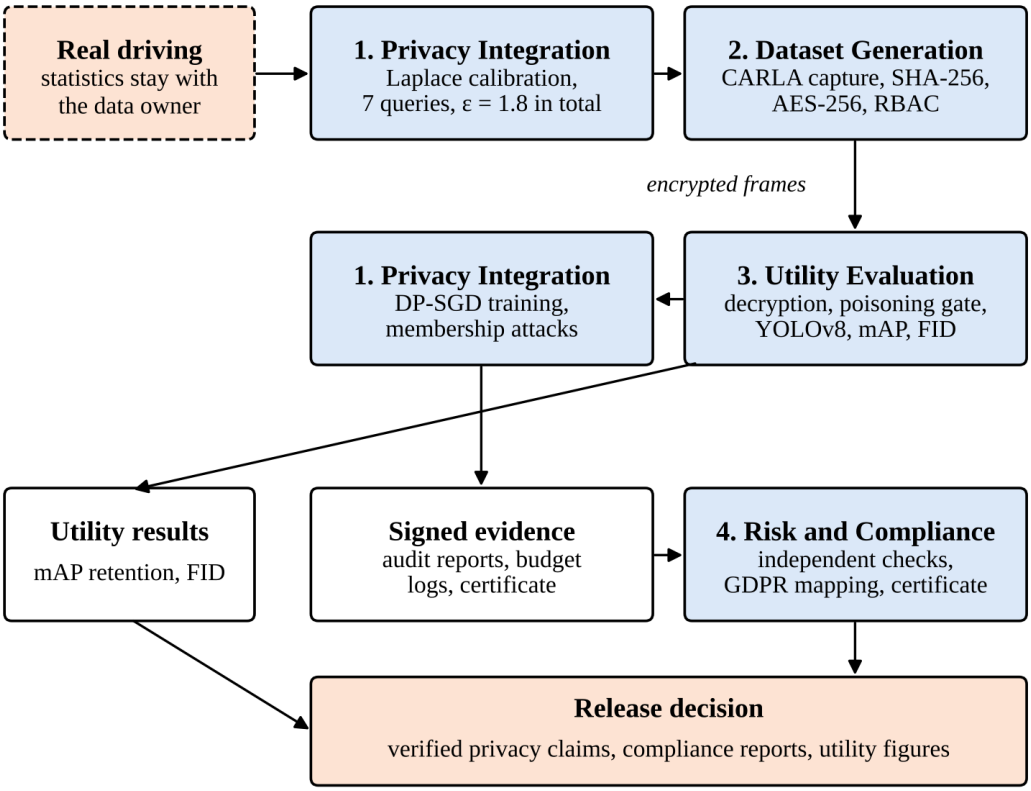


Figure 1.1: Overall system overview of the R26-CS-010 framework

1.3 Problem statement

Autonomous vehicle simulators are configured from real driving statistics. Current platforms compute and pass on these statistics without any privacy protection. They also give no formal guarantee for models trained on their output. Differentially private training has been applied to generative models. Most of this work used small benchmarks such as MNIST [8], [9]. Privacy budgets are often chosen for model accuracy. They are rarely checked against measured leakage [10], [11]. As a result, developers cannot show that their synthetic driving data does not expose real people. Nor can they give others a way to check such a claim.

1.4 Research gap

Earlier work leaves four gaps that matter for this component. First, simulator calibration is treated as a configuration task, so the real statistics behind it receive no protection [4]. Second, differentially private training has been tested mainly on benchmark datasets, not on driving images [12], [8]. Third, a formal budget ε is only a bound on paper. Few studies test it with a real attack against both private and non-private models [10], [11]. Fourth, privacy claims are rarely prepared so that a separate party can verify them. Section 2.5 explains these gaps in more detail.

1.5 Aim and objectives

The aim of this research is to integrate ε-differential privacy into the synthetic data pipeline. The privacy of the pipeline should then be measurable and verifiable. The project proposal set three measurable targets. The cumulative training budget should not exceed $\varepsilon = 10$. The success rate of a membership inference attack on the private models should stay at or below 55%. Random guessing gives 50%. Perception accuracy should stay within 15% of a model trained on real data. The specific objectives are:

1. To select a differentially private training library with per-sample gradients and Rényi differential privacy (RDP) accounting.
2. To protect the calibration step with the Laplace mechanism and keep the calibration budget below $\varepsilon = 2$.

3. To apply DP-SGD during training and stop training automatically at the budget limit.
4. To attack every trained model with a shadow-model membership inference attack, using a non-private baseline as a positive control.
5. To measure the trade-off between privacy and utility over three budgets and several random seeds.
6. To produce signed evidence that Risk and Compliance can verify independently.

1.6 Research questions or hypotheses

The main research question of this component is:

RQ: Can differential privacy at calibration and training stop membership inference on synthetic driving data at a measurable cost?

The question was divided into three sub-questions:

- **RQ1:** Can the seven calibration statistics be privatized before they reach the simulator, without changing how the simulator is used?
- **RQ2:** Does DP-SGD reduce membership inference to near random guessing when the same attack succeeds against a non-private model?
- **RQ3:** What does this protection cost in model utility, and does the cost change with the budget ε?

The working hypothesis was that the attack would succeed against the non-private baseline. Against every private model, it would fall to near 50%, with a measurable drop in test AUC.

1.7 Scope limitations and assumptions

The component covers two privacy points. These are the statistics that calibrate CARLA and the models trained on the generated frames. The component does not train a generative model on real records. CARLA renders frames directly from the privatized settings. For this reason, the Laplace step is where real information receives its formal guarantee. Applying DP-SGD to a classifier trained on the synthetic frames was a scope decision. It made it possible to test how ε relates to resistance against a real attack.

The work makes three assumptions. First, privacy was measured on a binary task: whether a frame contains a pedestrian or a cyclist. A simple task gives a clear membership signal. Second, the attacker was assumed to have query access to the model and data from the same distribution. Third, encryption, access control and file integrity were handled by the other components.

1.8 Individual contribution

The author designed and built the whole Privacy Integration component. The main contributions are:

- A calibration module that applies the Laplace mechanism to seven driving statistics within a total budget of $\varepsilon = 1.8$. It writes a public file and a private audit log.
- A DP training wrapper based on Opacus that logs ε after every step and stops training at a hard limit.
- A small convolutional classifier that trains under DP-SGD, with a label rule that removed label noise from the dataset.
- A shadow-model membership inference attack with a non-private positive control, and a second loss-threshold attack.
- A stress test on a deliberately overfitted model over five seeds, with bootstrap confidence intervals and paired one-sided t-tests.
- An implementation of a reference-calibrated attack.
- HMAC-signed audit reports and a privacy certificate, issued as a signed JSON file and a one-page PDF.
- The JSON contracts that connect the component to the other three components, and 135 unit tests.

1.9 Thesis structure

Chapter 2 reviews related work and builds the conceptual framework. Chapter 3 explains the research design, data, procedure and metrics. Chapter 4 presents the requirements and the design of the component. Chapter 5 covers implementation, integration and testing. Chapter 6 reports the results, and Chapter 7 discusses them. Chapter 8 considers commercialization and sustainability. Chapter 9 concludes the thesis.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review covered four topics that the component depends on. The first two were the theory of differential privacy and its use in deep learning. The other two were the privacy of synthetic data and simulators, and attacks that measure leakage. Most sources are peer-reviewed papers from security and machine learning venues. Examples are the IEEE Symposium on Security and Privacy, the USENIX Security Symposium, ACM CCS and CVPR. Preprints were used for tools that the component uses, such as Opacus. They were also used for recent work on autonomous vehicle privacy. Regulations and standards, such as the GDPR and the NIST Privacy Framework, defined what the evidence should support. A source was included if it met one of three conditions. It defined a mechanism used here, tested privacy on images or synthetic data, or measured privacy with an attack.

2.2 Domain and theoretical foundations

Dwork et al. [13], [14] defined differential privacy as a property of a randomized computation. Consider two datasets D and D′ that differ in one record, and any set of outputs S. A mechanism M satisfies ε-differential privacy if:

$$\Pr[M(D) \in S] \leq e^{\varepsilon} \cdot \Pr[M(D') \in S]$$

A smaller ε gives stronger protection but needs more noise. The relaxed form, (ε, δ)-differential privacy, allows this limit to fail with a small probability δ. This form is used for deep learning [15]. The Laplace mechanism answers a numerical query f by adding Laplace noise with scale $\Delta f / \varepsilon$. The sensitivity Δf is the largest change that one record can cause [13]. Two properties matter for this study. First, post-processing of a private output does not weaken the guarantee. Second, sequential composition applies when k mechanisms run on the same records. With budgets ε_1 to ε_k, they together satisfy $(\varepsilon_1 + ... + \varepsilon_k)$-differential privacy [15].

Abadi et al. [12] introduced differentially private stochastic gradient descent (DP-SGD). The gradient of each sample is clipped to a maximum L2 norm C. Gaussian noise with standard deviation σC is added to the sum. The result is averaged over a batch of size L:

$$\tilde{g} = (1/L) \cdot (\Sigma_i \, \mathrm{clip}(g_i, C) + N(0, \sigma^2 C^2 I))$$

Clipping limits how far one record can move the weights, and the noise hides the remaining effect. Mironov [16] introduced Rényi differential privacy (RDP). It generalizes the moments accountant of Abadi et al. Over many training steps, it composes more tightly than standard (ε, δ) composition.

Autonomous vehicle data adds its own privacy risks. Xie et al. [17] surveyed privacy risks in autonomous vehicles. They listed protection methods ranging from anonymization to private learning. The GDPR treats recordings of identifiable people as personal data. It also asks for data minimization and accountability [3]. Ganev [18] discussed when synthetic data can count as anonymous under such rules. He argued that formal privacy and attack testing matter for this question.

2.3 Related research

Private synthetic data. CARLA [4] is an open-source urban driving simulator. It produces camera, depth and segmentation data with automatic labels. Its traffic, pedestrians and weather can be set to match a real city. However, the CARLA paper does not discuss how the real statistics behind these settings should be protected. For learned generators, DP-GAN [8] adds Gaussian noise to the discriminator gradients of a generative adversarial network. DP-CGAN [9] adds label conditioning. These works were tested on MNIST and medical records, not on driving data. Stadler et al. [7] found that synthetic data without a formal mechanism did not reliably protect the source records. Private generators protected them better, but at a cost in utility. Protection must be placed where real records affect the output.

Membership inference. Shokri et al. [5] introduced the shadow-model attack. The attacker trains shadow models that copy the target. The attacker then records how these models respond to their own training records and to other records. An attack classifier is trained on these responses. Yeom et al. [19] linked membership leakage to

overfitting. They showed that the per-sample loss is a strong membership signal. Carlini et al. [6] showed that average measures can hide attacks that succeed on a few records. One such measure is the area under the ROC curve (AUC). They recommended reporting the true-positive rate at low false-positive rates.

Auditing DP in practice. Jayaraman and Evans [10] found that the leakage measured at a given ε varies widely between settings. Nasr et al. [11] used attacks to audit DP-SGD. They showed that a worst-case attacker can reach the formal bound. Realistic attackers fall well short of it. These studies agree on two points. A formal ε should be tested with an attack. A non-private model must also be attacked, to show that the attack works at all.

2.4 Existing systems products and techniques

Several tools and practices already address parts of the problem. Table 2.1 compares them with the needs of this component.

Table 2.1: Existing systems and techniques compared with the needs of the component

System or technique	What it provides	Limitation for this work
Face and plate blurring	Removes visible identifiers from images	Does not protect against membership inference at the model level
CARLA, AirSim and similar simulators	Labeled synthetic sensor data	No formal privacy guarantee; the calibration step is not protected [4]
DP-GAN and DP-CGAN	Private generative models [8], [9]	Tested on MNIST and medical data, not on driving data
Opacus	DP-SGD with per-sample gradients and an RDP accountant for PyTorch [20]	A training library only; it does not test the trained model with an attack
ML Privacy Meter	Membership inference audits of trained models [21]	Measures leakage but does not apply privacy or link results to a budget log

Opacus is the closest match for the training step. It implements DP-SGD and an RDP accountant for PyTorch. It also supports the Poisson sampling of batches that its accounting assumes [20]. Opacus does not allow batch normalization, because it mixes information across a batch. Private models commonly use group normalization instead

[22]. Auditing tools such as ML Privacy Meter [21] measure leakage after training. They do not apply the protection, and they do not link the leakage to a signed budget record.

2.5 Gap synthesis and conceptual framework

Table 2.2 summarizes the gaps that this component addresses and the approach taken for each.

Table 2.2: Research gaps addressed by the Privacy Integration component

Gap	Limitation of existing work	Approach in this component
Calibration privacy	Simulator statistics are computed from real data with no protection [4]	Laplace mechanism on every statistic before it reaches the simulator
DP on driving data	DP training was tested on benchmarks such as MNIST [12], [8]	DP-SGD on synthetic CARLA camera frames
Formal bound and attack	Budgets are chosen for accuracy and rarely checked with attacks [10], [11]	Shadow-model and loss-threshold attacks on every model, with a non-private control
Verification	Privacy claims are rarely prepared for third-party checks	Signed audit reports, budget logs and a certificate checked by another component

These gaps lead to the conceptual framework in Figure 2.1. Real information enters at two points, and each point has its own mechanism. The Laplace mechanism protects the calibration statistics, and DP-SGD protects the trained models. Each mechanism produces evidence in the form of a budget record and attack results. The attack results cover both private and non-private models. A signature protects each report, and a separate party checks the evidence. Privacy is claimed only when all three kinds of evidence agree.

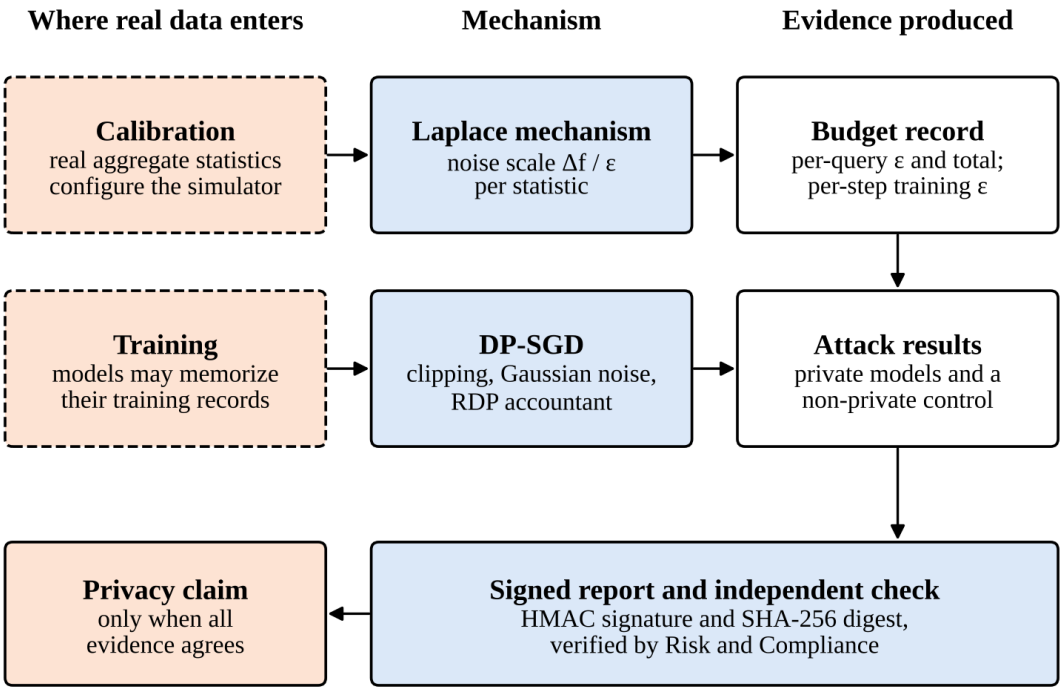


Figure 2.1: Conceptual framework of the Privacy Integration component

3 RESEARCH METHODOLOGY

3.1 Research design

The research followed a quantitative, experimental design inside a build-and-evaluate cycle. The component was first built as a working artifact. It was then tested in controlled experiments where the privacy setting was the only changing factor. The design had five parts:

- **Positive control.** A non-private baseline was trained and attacked in the same way as the private models. If the attack had failed against it, a private result near 50% would not prove privacy.
- **Manipulated factor.** The privacy budget took four levels: no DP, and DP-SGD with a target ε of 1, 3 and 8.
- **Repetition.** Each level was trained with three random seeds, so that differences could be tested for significance.
- **Stress test.** The baseline of the main study leaked only moderately. For this reason, a second experiment used a regime that makes the non-private model overfit. It used five seeds per level.
- **Independent check.** Risk and Compliance repeated the main checks with its own code and compared the results.

Figure 3.1 shows the research process from the problem to the verified results.

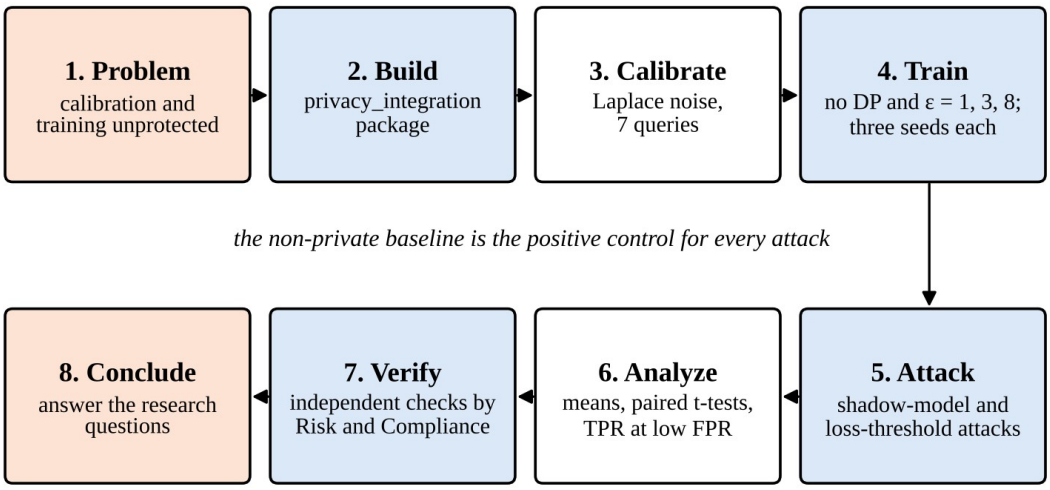


Figure 3.1: Research design and experimental procedure of the component

3.2 Data participants or study materials

No human participants took part in this study. The material was synthetic data and aggregate statistics.

Calibration statistics. The simulator needs seven aggregate statistics. These are vehicle and pedestrian density per kilometer, average vehicle and pedestrian speed, and intersections per kilometer. The weather and time-of-day distributions complete the set. These values were computed from the Waymo Open Dataset [2]. The computation ran in memory, and no raw record was written to disk.

Training frames. The training data was the five-town CARLA dataset of 10,000 frames. Dataset Generation captured these frames at 800 × 600 pixels. Every image was encrypted with AES-256 in CBC mode and listed in a SHA-256 manifest. The component decrypted each file and checked it against the manifest before use. The labels were the YOLO label files delivered with the dataset, and all 10,000 frames were used. The poisoning gate of Utility Evaluation later gave a FILTERED verdict on these labels. Its filtering was not applied to the results reported here.

Labels. Privacy was measured on a binary question: does the frame contain a pedestrian or a cyclist? A frame with neither was negative. A frame was positive when its largest pedestrian or cyclist covered at least 1,024 pixels (about 32 × 32). Frames with only smaller targets were excluded, because a negative label would have been wrong for them. This follows the KITTI benchmark, which leaves out objects too small to label reliably [1]. Table 3.1 summarizes the dataset after labeling and splitting.

Table 3.1: Dataset after label derivation and splitting

Item	Frames	Share
Positive (pedestrian or cyclist large enough to see)	796	8.0%
Negative (no pedestrian or cyclist)	7,052	70.5%
Excluded (only small targets)	2,152	21.5%
Training split (stratified)	5,493	70% of usable
Validation split	1,176	15% of usable
Test split	1,179	15% of usable

The 7,848 usable frames were split 70/15/15 with stratification. Images were resized to 448 × 336, which keeps the 4:3 aspect ratio. The training split was balanced by duplicating positive frames.

3.3 Experimental or study procedure

The procedure had six steps.

1. **Private calibration.** Each of the seven statistics was answered through the Laplace mechanism. The privatized file went to Dataset Generation. The noise log went only to Risk and Compliance. An earlier release ran every query at $\varepsilon = 1.8$. The corrected release splits a total of $\varepsilon = 1.8$ evenly, at $\varepsilon \approx 0.2571$ per query. The dataset used here was captured from the corrected release (Sections 6.1 and 6.3).
2. **Data preparation.** The encrypted frames were decrypted and checked against the manifest. They were then labeled with the rule in Section 3.2 and split with a fixed seed.
3. **Training.** A non-private baseline and three private configurations were each trained with seeds 42, 43 and 44. This gave twelve models.
4. **Attack.** Every model was attacked with the shadow-model attack and then with the loss-threshold attack.
5. **Stress test.** The baseline and the three private configurations were trained again on 500 members for 60 epochs. Seeds 42 to 46 were used, and each model faced the loss-threshold attack.
6. **Evidence.** Every run produced a signed audit report. A certificate combined all runs and the significance tests.

Table 3.2 lists the four configurations. Opacus took the target ε, $\delta = 10^{-5}$, the number of epochs and the sampling rate as inputs. It computed the noise multiplier σ from them.

Table 3.2: Experimental configurations

Configuration	Target ε	Mechanism	Seeds
No DP (baseline)	none	Adam, 15 epochs	42, 43, 44
Config A	1.0	DP-SGD, 10 epochs, C = 1.0, $\delta = 10^{-5}$, σ = 0.977	42, 43, 44
Config B	3.0	DP-SGD, 10 epochs, C = 1.0, $\delta = 10^{-5}$, σ = 0.665	42, 43, 44
Config C	8.0	DP-SGD, 10 epochs, C = 1.0, $\delta = 10^{-5}$, σ = 0.507	42, 43, 44

Per-sample gradients were clipped to C = 1.0. The private models used SGD with momentum, a learning rate of 10^{-3} and a batch size of 32. The baseline used the same model and data with Adam, a learning rate of 10^{-4} and 15 epochs. The RDP accountant was read after every step. Its cumulative ε was written to a budget log. Training would have stopped with an error if ε had reached the hard limit of 10.

Table 3.3 gives the settings of the stress test. Each target was trained on 500 member frames and evaluated against 500 non-member frames. With few members and many epochs, the non-private model memorizes its training set. This gives the attack a clear signal to find. The private models were trained in the same regime, so any drop in attack AUC comes from DP-SGD.

Table 3.3: Settings of the stress test

Setting	Value
Members and non-members	500 and 500 frames
Epochs and batch size	60 epochs, batch size 32
Non-private optimizer	Adam, learning rate 10^{-3}
Private optimizer	SGD with momentum, learning rate 10^{-3}, C = 1.0, $\delta = 10^{-5}$
Budgets	ε = 1, 3 and 8
Seeds	42, 43, 44, 45 and 46
Attack	Loss threshold, with 1,000 bootstrap resamples for a 95% confidence interval

The shadow-model attack trained a shadow model on half of the data with the privacy setting of the target [5]. Two features were recorded for each of its members and non-members. These were the confidence and the per-sample loss [19]. A small neural network learned to separate members from non-members with these features. It was then applied to the target. The members were 1,179 training images of the target. The non-members were its 1,179 test images. With equal set sizes, random guessing scores exactly 50%. The loss-threshold attack needed no shadow model. It treated a lower loss on the target model as a sign of membership and used the same two sets.

3.4 Measures and evaluation metrics

Table 3.4 lists the measures used in the evaluation.

Table 3.4: Measures and evaluation metrics

Measure	Definition	Purpose
Achieved ε	Final value of the RDP accountant at $\delta = 10^{-5}$	Formal privacy cost of training
Attack success rate	Share of members and non-members classified correctly	Main proposal target (≤ 55%)
Attack AUC	Area under the ROC curve of the attack	Leakage over all thresholds (0.5 = chance)
TPR at low FPR	True-positive rate at 1% and 5% false-positive rate [6]	Leakage on the most exposed records
Loss gap	Mean loss of non-members minus mean loss of members	Size of the overfitting signal
Test AUC	AUC of the classifier on the test split	Utility of the trained model
Advantage removed	Drop in attack AUC as a share of the baseline advantage over 0.5	Size of the protection in the stress test

Test AUC measured utility instead of accuracy. The test split keeps the natural positive rate of about 10%. A model that always predicts "no pedestrian" would reach a high accuracy and still be useless.

3.5 Data analysis

Each measure was reported as the mean and standard deviation over the seeds. ROC curves and AUC values were computed with scikit-learn. Each private configuration

was compared with the baseline trained with the same seed. In the main study, a two-sided paired t-test was applied to the drop in test AUC and in attack AUC. A result was significant at the 0.05 level. With three seeds, these tests can separate a private model from the baseline. They have little power to separate the three budgets from each other.

The stress test used two standard rules. The first version of the test used stricter cut-offs of its own. These were replaced by the standard rules after the first run. The certificate reports both verdicts (Section 7.5). The positive control passes only if every non-private run has a 95% bootstrap interval fully above 0.5. The reduction is significant only if a one-sided paired t-test over five seeds gives $p < 0.05$ for every budget. Paired one-sided t-tests between neighboring budgets were also reported. They check whether a smaller ε gives stronger protection.

For the calibration step, the analysis checked three things. First, the Town01 simulator configuration was compared field by field with the privatized file. Second, the total calibration budget was computed under sequential composition, because all seven queries read the same records. Third, Risk and Compliance compared the published values with the source values. This showed whether the noise had the size that the Laplace mechanism requires.

3.6 Ethics privacy security and responsible AI

All images in this work were generated inside the CARLA simulator. No real person, vehicle or place was recorded by the project. However, the calibration statistics came from the Waymo Open Dataset [2]. This dataset was recorded on public roads. Only aggregate values were kept, not individual records (Section 3.2). The calibration audit log records the noise drawn for each query. Subtracting this noise from a published value gives back the source value. For this reason, the audit log was shared only with Risk and Compliance and was never published.

The AES dataset key and the HMAC signing key were kept separate. Both keys were loaded from a local .env file that is excluded from version control. Before the full experiment, a code review found both keys in source files of a repository due for publication. Both keys were replaced. A regression test now fails if an old key appears in the source again.

Every privacy claim in this thesis is tied to an attack result against a positive control. Known weaknesses are recorded in the signed certificate. The component does not claim legal compliance. It produces evidence that a data controller can use when making that decision.

3.7 Reproducibility and research artefacts

Every source of randomness was seeded. Repeating a run with a fixed seed gave the same loss and the same test AUC. All settings were kept in one file, config.yaml. Conflicting values were rejected when the package loaded. Results were saved in a separate folder for each seed, so a new run never overwrote an earlier one. The software versions were fixed at Python 3.11, PyTorch 2.5.1 and Opacus 1.4.1.

The main research artefacts are the privacy_integration package and the twelve trained models of the main study. Other artefacts are the twenty stress-test results and the per-step budget logs. The signed audit reports and the privacy certificate complete the set. Each report is stored with its SHA-256 digest, so a later change to any artefact can be detected. All twenty stress-test result files are signed with HMAC-SHA256.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements came from three sources: the project proposal, the needs of the other three components and the stakeholders. The proposal named four stakeholder groups. Autonomous vehicle developers need training data with an auditable privacy record. Road users need their data to stay private. Data protection regulators need evidence they can check. Researchers want to share datasets safely. Table 4.1 lists the functional requirements.

Table 4.1: Functional requirements of the Privacy Integration component

ID	Requirement	Source
FR-01	Every private training run stays within its target ε	Proposal target
FR-02	Per-sample gradients are clipped and noise is added at every step	Definition of DP-SGD [12]
FR-03	Training stops automatically if ε reaches the hard limit of 10	Proposal target
FR-04	Noise is added to the statistics before they reach the simulator	Calibration gap
FR-05	Every run produces a signed report that fails verification if changed	Needs of Risk and Compliance
FR-06	The cumulative ε is logged after every training step	Needs of Risk and Compliance
FR-07	The calibration queries together stay within the declared total budget	Sequential composition [15]

The non-functional requirements were as follows:

- **Security.** Keys are never written in the source code. Files that could reveal source values are never published.
- **Verifiability.** Another party can check every claim from the files alone, without repeating the training.
- **Reproducibility.** A run repeated with the same seed gives the same result.
- **Interoperability.** Every file exchanged with another component follows a JSON Schema contract.

- **Resource limits.** Training and attacks run on a laptop GPU with 4 GB of memory.

4.2 Component architecture within the complete solution

Figure 4.1 shows where the component sits in the group pipeline. It works at two points. Before the simulator, it receives aggregate statistics on the data owner's machine. Only privatized values go to Dataset Generation. After the simulator, it trains models on the synthetic frames with DP-SGD and attacks them. It then sends signed evidence to Risk and Compliance. In pipeline mode, the poisoning gate of Utility Evaluation supplies the clean labels for training.

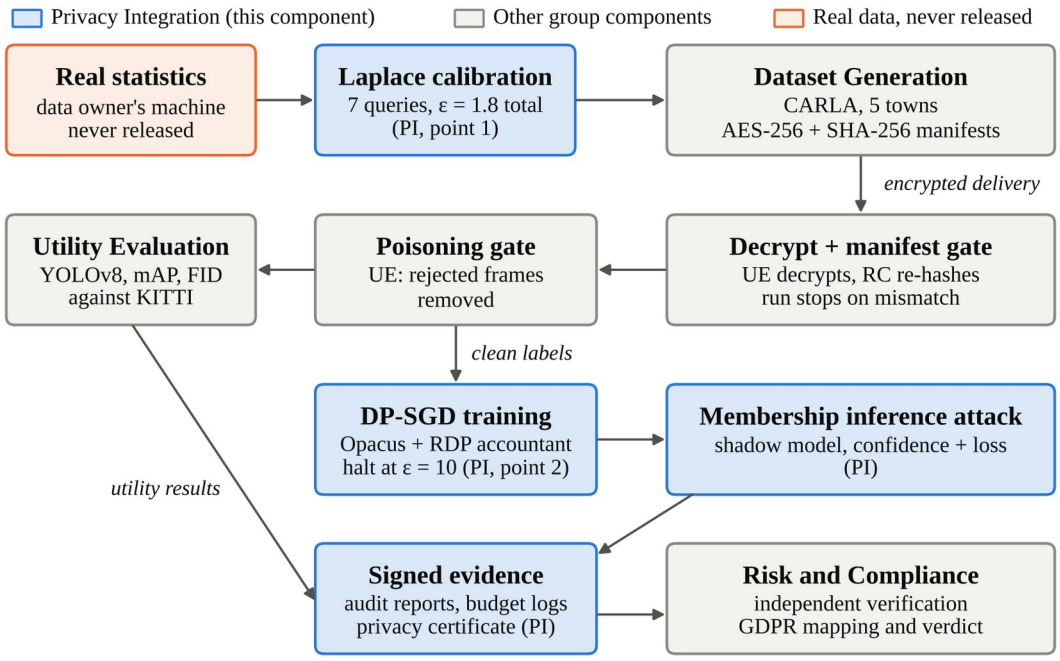


Figure 4.1: Group pipeline architecture and the two points where Privacy Integration acts

This position gives the component a clear boundary. The file calibration_stats.json is the only path from the source statistics to the simulator. The signed reports are the only path from the training runs to the auditor. As a result, both paths can be checked in one place.

4.3 Design alternatives and decisions

Table 4.2 lists the main design choices, the alternatives considered and the reason for each choice.

Table 4.2: Design alternatives and decisions

Decision	Alternative	Reason for the choice
Privatize the calibration statistics	Train a private generator on real records	CARLA renders frames from its settings, so the statistics are the only point where real data enters the generator
Laplace mechanism for calibration	Gaussian mechanism	Gives pure ε-differential privacy for a small number of numeric queries [13]
Opacus for DP-SGD	TensorFlow Privacy	Computes per-sample gradients without model changes, provides RDP accounting and works with the PyTorch code of the group [20]
RDP accountant	Basic sequential composition	Composes more tightly over thousands of training steps [16]
Group normalization	Batch normalization	Opacus does not allow batch normalization, which mixes samples across a batch [22]
SGD with momentum for private runs	Adam	Adam was unstable under DP-SGD in the early runs
Duplicating positive frames	Weighted sampler or class weights	Opacus replaces a weighted sampler, and class weights did not work well with per-sample clipping
Shadow-model and loss-threshold attacks	A single attack	Two different attacks reduce the chance that one weak attack hides leakage [5], [19]
HMAC-SHA256 signatures	Public-key signatures	Simple to verify with a shared key; public-key signatures are noted as future work

The first decision shaped the rest of the design. The final pipeline has no generative model trained on real records. CARLA renders the frames directly from the privatized settings. For this reason, DP-SGD was applied to a classifier trained on the synthetic frames. This choice made it possible to test how ε relates to resistance against a real attack. That link was one of the gaps in Section 2.5.

4.4 Detailed design

The component is an installable Python package called privacy_integration. It has six modules. Figure 4.2 shows how data moves between them.

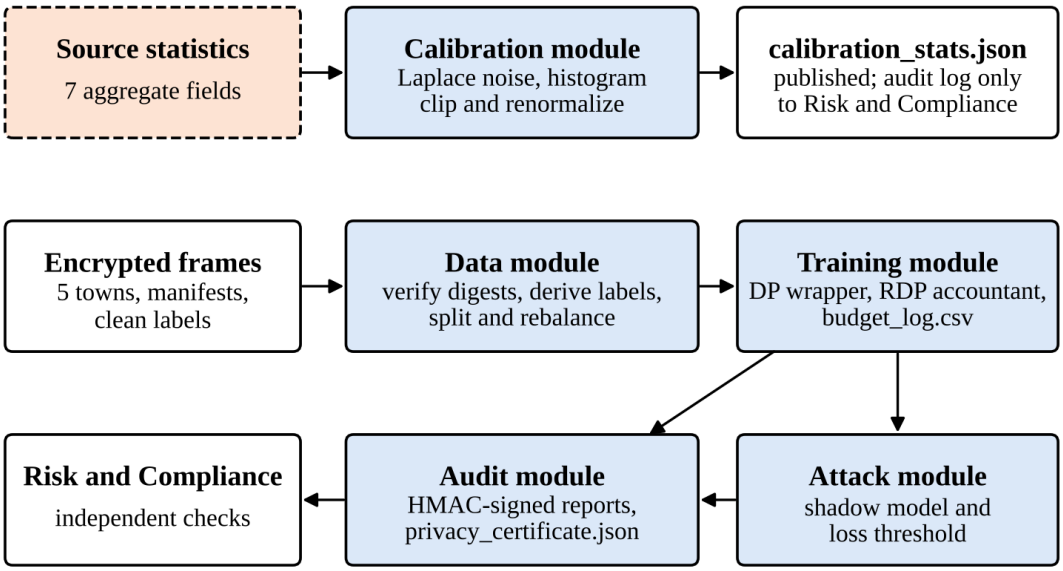


Figure 4.2: Internal data flow of the Privacy Integration component

Calibration module. Each statistic is one query answered through the Laplace mechanism. The module takes a total budget, here ε = 1.8. It splits this budget evenly over the seven queries. Each query then runs at ε = 1.8 / 7 ≈ 0.2571. Each query also stays below a configured per-query limit of 2.0. The sensitivity is 2.0 for vehicle speed and 1.0 for the other queries. For the two distributions, noise is added to every bin. Negative bins are then set to zero, and the bins are scaled to sum to one. This is post-processing, so it does not weaken the guarantee. Table 4.3 lists the queries.

Table 4.3: Calibration queries, sensitivities and Laplace noise scale at ε ≈ 0.2571 per query (total ε = 1.8)

Query	Type	Sensitivity Δf	Noise scale b
Vehicle density per km	Scalar	1.0	3.889
Pedestrian density per km	Scalar	1.0	3.889
Average vehicle speed (km/h)	Scalar	2.0	7.778
Average pedestrian speed (km/h)	Scalar	1.0	3.889

Query	Type	Sensitivity Δf	Noise scale b
Intersections per km	Scalar	1.0	3.889
Weather distribution (4 bins)	Histogram	1.0	3.889 per bin
Time-of-day distribution (4 bins)	Histogram	1.0	3.889 per bin

All seven queries read the same records. Under sequential composition, their costs add up to 7 × 0.2571 = 1.8. The module reports this total as calibration_epsilon_spent. Earlier versions ran every query at ε = 1.8 and reported 1.8. The composed total, however, was 12.6 (Section 6.3). The per-query mode is still available for comparison, but the total mode is the default. A noised scalar can fall outside its natural range. For example, the published intersection count is negative. Mapping such a value into range during simulator configuration is post-processing and costs no budget. The module writes two files. The file calibration_stats.json holds the privatized statistics. The file calibration_audit_log.json records the noise for each query and goes only to Risk and Compliance.

Data module. This module decrypts each frame and checks it against the SHA-256 manifest. It then applies the label rule and creates the stratified splits.

Model module. The classifier is a small convolutional network, listed in Table 4.4. Group normalization replaces batch normalization. Max pooling and limited downsampling help small pedestrians reach the output.

Table 4.4: Presence classifier architecture (input 3 × 336 × 448)

Stage	Operation	Output channels
Block 1	3 × 3 convolution, stride 2, GroupNorm, ReLU	32
Block 2	3 × 3 convolution, stride 2, GroupNorm, ReLU	64
Block 3	3 × 3 convolution, stride 2, GroupNorm, ReLU	128
Block 4	3 × 3 convolution, stride 1, GroupNorm, ReLU	128
Pooling	Adaptive max pooling to 1 × 1	128
Classifier	Linear 128 to 64, ReLU, Linear 64 to 1	1 (logit)

Training module. This module holds the DP training wrapper, the budget accountant and the non-private baseline.

Attack module. This module trains the shadow models, builds the attack classifier and runs the loss-threshold attack.

Audit module. This module signs the reports and builds the privacy certificate. After every private run, it writes an audit report. The report holds the final ε, C, σ, δ, the limit and the number of steps. It is signed with HMAC-SHA256, and its SHA-256 digest is stored next to it. For the baseline, a signed declaration states that no privacy mechanism was applied. The file privacy_certificate.json combines the calibration budget, every configuration, the stress test and the utility cost. It also lists the declared limitations and is signed in the same way. A one-page PDF summarizes the certificate. It prints the SHA-256 digest, the HMAC signature and a QR code of the digest. One line is left blank for a handwritten signature. A verify command recomputes both values. It fails if the JSON file or the PDF has changed.

4.5 Standards and professional practice

The design follows the standards and practices listed below.

- **Data protection by design.** Privacy is applied where real data enters the pipeline, as GDPR Article 25 requires [3]. Only privatized statistics leave the data owner. This supports the data minimization principle of Article 5(1)(c).
- **Accountability.** Signed reports and budget logs support the accountability duty of GDPR Article 5(2) [3].
- **Risk frameworks.** The evidence supports the Control and Protect functions of the NIST Privacy Framework [23]. It also supports the privacy and security goals of the NIST AI Risk Management Framework [24].
- **Cryptographic standards.** Digests use SHA-256 as defined in FIPS 180-4 [25].
- **Engineering practice.** Secrets were kept out of version control. Code was reviewed before the full experiment. Unit tests ran on every push through continuous integration.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

All training and attack runs used a laptop with an NVIDIA RTX 3050 Ti GPU (4 GB) running Windows 11. Table 5.1 lists the main tools and their purpose.

Table 5.1: Development environment and technology stack

Tool	Purpose
Python 3.11	Main programming language
PyTorch 2.5.1	Building and training the convolutional classifier
Opacus 1.4.1	DP-SGD with per-sample gradients and an RDP accountant
NumPy and SciPy	Laplace noise, statistics and paired t-tests
scikit-learn	ROC curves and AUC for the attacks
PyCryptodome	Decrypting the dataset received from Dataset Generation
Python hmac and hashlib	HMAC-SHA256 signatures and SHA-256 digests
ReportLab	One-page certificate PDF with a QR code of the digest
JSON Schema	Checking every file exchanged with other components
pytest and GitHub Actions	135 unit tests and continuous integration

5.2 Component implementation

Each module of the package has a command-line entry point. This means every stage can be run and checked on its own. All settings are read from config.yaml when the package loads.

Training wrapper. The DPTrainingWrapper class attaches an Opacus PrivacyEngine to the model, optimizer and data loader. It calls make_private_with_epsilon to find the noise multiplier σ for the target ε. After every step, it reads the accountant and adds a row to the budget log. If the cumulative ε reaches the limit, it raises BudgetExceededError and stops the loop. Opacus secure mode was turned off. Secure mode needs the torchcsprng library. This library is no longer actively maintained and requires PyTorch 1.8.1. It cannot be used with PyTorch 2.5 on any operating system. The certificate lists this as a limitation.

Audit and certificate. The CertificateSigner class signs each report. The key is read from the project .env file, and verification uses a constant-time comparison. The

build_certificate script reads the results of every run and computes the paired t-tests. It then writes a signed privacy_certificate.json.

Stress test and calibrated attack. The pi-stress command trains the stress-test targets and attacks them with the loss-threshold attack. It writes one file for each seed and configuration. It also writes a summary with the bootstrap intervals, the paired tests and the verdict. The seeds were run at different times. For this reason, pi-stress-rescore rebuilds the summary from all saved files without retraining. The pi-stress-calibrated command implements a reference-calibrated attack in the style of Carlini et al. [6]. Four non-private reference models per seed are trained on frames that are never attacked. They estimate how hard each frame is. The attack then scores how much better the target does on a frame than the references expect. It is estimated to take about 2.5 hours per seed on the laptop GPU.

Implementation challenges. In the first runs on real frames, both private and non-private models predicted the majority class for every input. A search over learning rates, optimizers, batch sizes and imbalance methods gave three findings. Adam was unstable under DP-SGD, so private training was switched to SGD with momentum. Loss-based class weighting did not work well with per-sample clipping. Opacus also replaces a weighted sampler, so balancing had to be done by duplicating frames.

None of these changes fixed the problem. A label check then examined the positive frames. In 62.3% of them, the largest pedestrian was smaller than about 8 × 8 pixels after resizing. This meant most positives were label noise. The frames were also being stretched into a 256 × 256 square. The aspect ratio was then kept, and frames with only small targets were excluded. After these changes, the baseline test AUC rose from about 0.50 to 0.84. The earlier failure came from data preparation, not from differential privacy.

5.3 Integration with the other components

A pipeline runner connects the four components in order. Before any training starts, Risk and Compliance re-hashes the decrypted dataset. The poisoning gate of Utility Evaluation then produces the clean labels. The component has three pipeline hooks. All three are off by default, so standalone results do not change. R26_OUTPUTS_DIR

moves the outputs into the run workspace. The --labels-dir option trains on the labels that passed the poisoning gate and skips rejected frames. The --utility-results option adds the results of Utility Evaluation to the certificate. Every file passed between components is checked against a JSON Schema contract. A missing or wrongly typed field stops the run at that stage. Table 5.2 lists the files that the component exchanges.

Table 5.2: Files exchanged by the Privacy Integration component

File	Direction	Purpose
calibration_stats.json	To Dataset Generation and Risk and Compliance	Privatized statistics that configure CARLA
calibration_audit_log.json	To Risk and Compliance only	Noise per query for the reconstruction and composition checks; never published
Encrypted towns and manifests	From Dataset Generation	Training frames and integrity digests
clean_labels/	From Utility Evaluation	Labels that passed the poisoning gate (pipeline mode)
utility_results.json	From Utility Evaluation	Detection utility for the utility section of the certificate
audit_report.json and audit_certificate.json	To Risk and Compliance	Signed evidence for each run
summary.json	To Utility Evaluation	Attack AUC and test AUC per configuration for the joint figure
privacy_certificate.json	To the release folder	Combined and signed privacy verdict

The AES dataset key and the HMAC signing key traveled separately. Dataset Generation shared the AES key so that the frames could be decrypted. Risk and Compliance received the HMAC key so that it could verify the reports.

5.4 Verification and software testing

The component was tested at three levels. Unit tests checked each module on its own. They cover the halt at the budget limit and the rejection of a changed report. Other tests cover the calibration budget, the stress-test verdict rules and the certificate PDF. A regression test fails if an old key appears in the source. All 135 unit tests passed. Shared library and pipeline tests checked the JSON contracts between the components. An

end-to-end test ran the whole pipeline. GitHub Actions ran the tests of every component and the end-to-end test on each push.

The functional requirements were then checked with the full experiment and the stress test. The full experiment trained twelve models and attacked each one. Table 5.3 lists how each requirement was verified.

Table 5.3: Verification of the functional requirements

ID	Requirement	Verification	Status
FR-01	Runs stay within target ε	ε = 0.991, 3.000 and 7.995	Passed
FR-02	Clipping and noise every step	C = 1.0, σ = 0.977 to 0.507	Passed
FR-03	Training halts at the limit	Halts in a low-limit test	Passed
FR-04	Noise added before simulator	Town01 matched privatized file	Passed
FR-05	Signed report for every run	Altered copy is rejected	Passed
FR-06	Per-step ε is logged	3,090 rows per run	Passed
FR-07	Calibration within the total	7 × 0.2571 = 1.8; composition check passed	Passed

Repeating a run with a fixed seed gave the same loss and the same test AUC. This confirmed that the runs were reproducible.

5.5 Deployment and operation

The component is installed as a Python package and run stage by stage from the command line. In the intended deployment, an operator first runs the calibration stage on the data owner's machine. Only calibration_stats.json leaves that machine for the simulator. The audit log goes to the auditor through a separate channel. The training and attack stages then run on a GPU machine. This machine has access to the encrypted dataset, the clean labels and the AES key. Finally, the certificate stage collects the results of all runs and signs the combined verdict.

The keys are supplied through environment variables or a local .env file. This lets the same code run on any machine without changes. Each run writes to its own folder, which makes it safe to repeat an experiment or add a new seed. The budget log, the signed report and its digest stay together in that folder. An auditor can check any run from its folder alone.

6 RESULTS AND VALIDATION

6.1 Research results

Privacy and utility. Table 6.1 gives the mean over the three seeds for each configuration. Figure 6.1 plots the two main measures.

Table 6.1: Privacy and utility by configuration (mean ± standard deviation over three seeds)

Configuration	Target ε	Test AUC	Attack AUC	Attack success	Loss gap
No DP	none	0.829 ± 0.009	0.639 ± 0.013	56.7%	0.360
Config A	1.0	0.621 ± 0.045	0.504 ± 0.002	50.3%	0.001
Config B	3.0	0.655 ± 0.035	0.515 ± 0.002	51.6%	0.013
Config C	8.0	0.668 ± 0.020	0.506 ± 0.007	50.8%	0.026

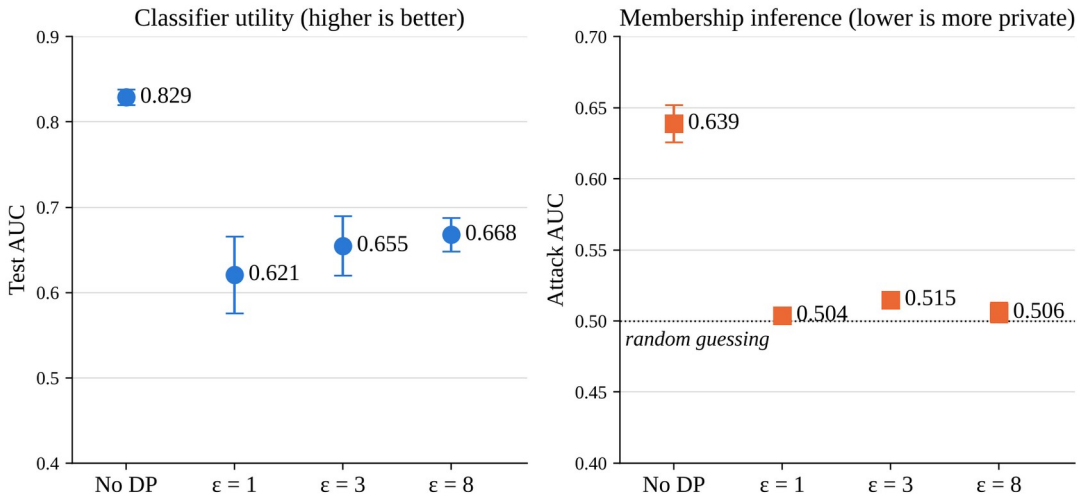


Figure 6.1: Test AUC and attack AUC by configuration (mean and one standard deviation over three seeds)

The attack worked against the non-private baseline, with a mean AUC of 0.639 and a success rate of 56.7%. On training images, the baseline loss was 0.36 lower than on unseen images. The attack uses this overfitting signal. Against every private configuration, the attack AUC fell to between 0.504 and 0.515. The success rate fell to between 50.3% and 51.6%, close to random guessing. The ranges of single runs did not overlap. The nine private runs had an attack AUC between 0.499 and 0.516. The three baseline runs had an attack AUC between 0.625 and 0.650. The cost was a fall in test AUC from 0.829 to between 0.621 and 0.668.

Each private configuration was compared with the baseline trained with the same seed. Two-sided paired t-tests showed that both drops were significant at the 0.05 level. This held for every private configuration. Table 6.2 gives the details. With only three seeds, the differences between the three private configurations were not significant.

Table 6.2: Two-sided paired t-tests of each private configuration against the baseline (three seeds)

Configuration	Drop in test AUC	p-value	Drop in attack AUC	p-value
Config A	0.208	0.0099	0.134	0.0025
Config B	0.174	0.0084	0.124	0.0032
Config C	0.160	0.0031	0.132	0.0031

Loss-threshold attack. Every saved model was also tested with a loss-threshold attack [19]. It used the same 1,179 members and 1,179 non-members. Every run reproduced the test AUC recorded at training, which confirmed that the splits matched. Following Carlini et al. [6], Table 6.3 reports the true-positive rate (TPR) at a low false-positive rate (FPR). Figure 6.2 compares the two attacks.

Table 6.3: Loss-threshold attack (mean ± standard deviation over three seeds; random guessing gives an AUC of 0.5 and a TPR equal to the FPR)

Configuration	Attack AUC	TPR at 1% FPR	TPR at 5% FPR
No DP	0.610 ± 0.026	0.007 ± 0.007	0.057 ± 0.012
Config A	0.501 ± 0.007	0.015 ± 0.002	0.053 ± 0.011
Config B	0.503 ± 0.012	0.016 ± 0.003	0.055 ± 0.013
Config C	0.506 ± 0.012	0.014 ± 0.006	0.053 ± 0.009

The second attack confirmed the main result. Its AUC against the baseline was 0.610. For every private run, it fell to between 0.491 and 0.518. The paired drop of 0.10 to 0.11 was significant for each private configuration ($p = 0.025$, 0.033 and 0.035). At low false-positive rates, however, the attack performed at chance level for every configuration. This includes the baseline. These values are approximate, because a 1% FPR allows only about 12 false alarms. They still show that the baseline leakage was a small average shift in loss across many images. No single frame could be identified with high confidence, even without DP.

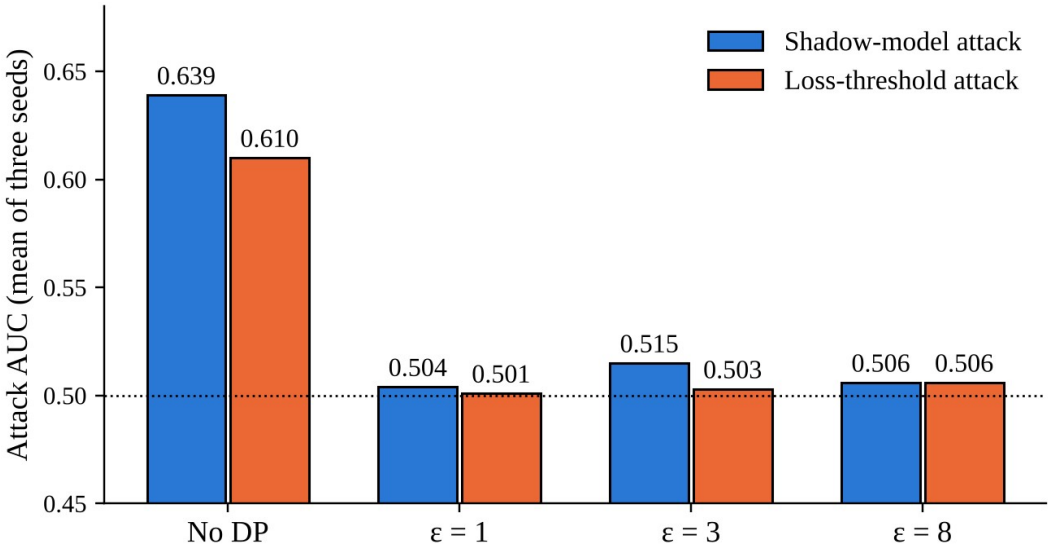


Figure 6.2: Attack AUC of the shadow-model and loss-threshold attacks

Stress test. Table 6.4 and Figure 6.3 give the results of the stress test. In this regime, the non-private model leaked clearly. Its attack AUC averaged 0.628 over the five seeds. The 95% bootstrap interval of every seed lay fully above 0.5, so the positive control passed. DP-SGD reduced the attack at every budget, and every drop was significant. At ε = 1, the attack AUC fell to 0.501. This removed 99.5% of the attacker's advantage over random guessing. At ε = 3 and ε = 8, 79.4% and 63.0% of the advantage was removed.

Table 6.4: Stress test on an overfitted model (mean over five seeds; one-sided paired t-tests against the non-private model)

Configuration	Attack AUC	Drop in attack AUC	p-value	Advantage removed
No DP	0.628			
ε = 1	0.501	0.128	0.00004	99.5%
ε = 3	0.526	0.102	0.00006	79.4%
ε = 8	0.547	0.081	0.001	63.0%

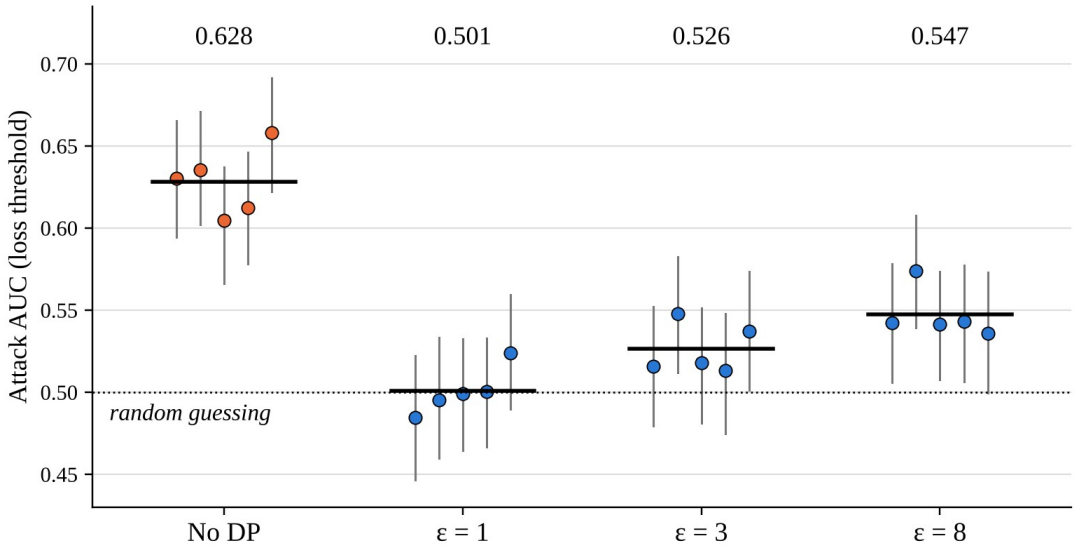


Figure 6.3: Stress test: attack AUC of every seed with its 95% bootstrap interval; the black line marks the mean of each configuration

The stress test also separated the budgets, which the main study could not do. A smaller ε gave a lower attack AUC, and both steps were significant. The difference between ε = 1 and ε = 3 was 0.026 ($p = 0.013$). The difference between ε = 3 and ε = 8 was 0.021 ($p = 0.010$). Once enough leakage was present, the formal order of the budgets appeared in the measured leakage. The models in this regime had low utility by design. They were trained on only 500 frames. Their mean test AUC was 0.656 without DP and 0.539 to 0.562 with DP.

Calibration. The calibration profile in the Town01 configuration matched the privatized file field by field. This confirms that the simulator used the privatized values. Table 6.5 shows the two calibration releases. The first release ran each query at ε = 1.8. The corrected release splits a total of ε = 1.8 over the seven queries. As a result, each query receives about seven times more noise. The dataset was captured again from the corrected release. All results in this thesis come from this capture. The source values and the noise drawn are kept only in the audit logs.

Table 6.5: Privatized calibration statistics of the two releases

Statistic	Earlier release	Corrected release (used for capture)
Vehicle density per km	2.5139	5.1602
Pedestrian density per km	0.9153	0.4806
Average vehicle speed (km/h)	22.0135	30.4335
Average pedestrian speed (km/h)	0.2789	1.9526
Intersections per km	0.2796	-5.2851
Weather: clear, cloudy, rain, fog	0.624, 0.135, 0.241, 0.000	0.650, 0.156, 0.195, 0.000
Time of day: morning, afternoon, evening, night	0.144, 0.115, 0.575, 0.167	0.000, 0.000, 0.837, 0.163
ε per query	1.8	0.2571
Total ε under sequential composition	12.6	1.8

The larger noise of the corrected release is visible in the scalar values. The intersection count became negative. This is a valid output of the mechanism. It is mapped into range when the simulator is configured. Section 6.4 reports how Risk and Compliance checked this noise.

6.2 Product and technical results

The component was delivered as a working package. It runs the whole privacy workflow from calibration to the signed certificate. All seven functional requirements passed (Table 5.3). Every private run stayed within its target. The final values were ε = 0.991, 3.000 and 7.995 for Config A, B and C. No run reached the hard limit of 10. A separate low-limit test confirmed the halt itself. Each run logged 3,090 rows of cumulative ε, one for each training step. The corrected calibration release composes to exactly ε = 1.8.

The audit trail worked as designed. Every run produced a signed report, and a changed copy of a report was rejected. The final privacy certificate is one signed file. It combines the twelve runs, the paired t-tests, the stress test and the utility results. It also lists the declared limitations. A separate record stores the SHA-256 digest of its PDF. The certificate verdict was PASS for both privacy enforcement and empirical

validation. The package met its non-functional requirements as well. No key remained in the source code, and runs were reproducible with a fixed seed. Repeating a run gave the same per-epoch loss and test AUC for Config A and the baseline. This was confirmed at two seeds. The whole experiment ran on a laptop GPU with 4 GB of memory.

6.3 User or stakeholder validation

The component had no external end users during the project. Validation came instead from three groups that used or reviewed its outputs.

Supervisors. The supervisor and co-supervisor reviewed the work at the end of each sprint. Their early advice narrowed the privacy work to two clear points, calibration and training. Both could be built and tested within the project timeline.

Risk and Compliance as the user of the evidence. Risk and Compliance used the signed reports, budget logs and calibration audit log as its inputs. It verified all of them without help from the author. This met the requirement that another party can check every claim. Its composition check also found an error. Earlier versions of the code ran every calibration query at $\varepsilon = 1.8$ and reported 1.8. This was the budget of one query only. Under sequential composition, the correct total was $7 \times 1.8 = 12.6$. The calibration module was changed to take a total budget and split it over the queries. The corrected release composes to 1.8, and the check now passes. Each query now runs at a smaller ε. The corrected release therefore has more noise and different values (Table 6.5). The dataset was then captured again from the corrected release (Section 6.1).

Progress review panel. At the second progress review, the panel commented that the reduction in attacks was not significant, even with differential privacy. In statistical terms, the drop in the main study was significant (Table 6.2). In absolute terms, however, it was small. The baseline itself leaked only moderately, with a success rate of 56.7%. At a 1% FPR, it was no better than chance. This left DP with little leakage to remove. In response, the stress test was designed and run. Against a baseline that leaks clearly, DP-SGD removed 99.5% of the attacker's advantage at $\varepsilon = 1$. Every reduction was significant over five seeds, and smaller budgets gave significantly stronger protection (Table 6.4).

6.4 Integration results

The integration was checked from both sides of each interface. Dataset Generation configured the simulator from calibration_stats.json. The Town01 profile matched the privatized file field by field. Utility Evaluation used summary.json to draw the joint privacy and utility figures of the group. Its utility results were also added to the certificate. Risk and Compliance checked the outputs of the component with its own code. Table 6.6 lists the results.

Table 6.6: Independent verification of the component by Risk and Compliance

Check	What was tested	Result	Outcome
Report signatures	Every signed audit report verifies	12 of 12 reports verified	Passed
Tamper test	A changed report is detected	ε changed to 0.01 was caught	Passed
Budget audit	Reported ε matches the training log	Identical after 3,090 steps	Passed
Budget ceiling	ε never decreases or exceeds 10	Peaks of 0.99, 3.00 and 7.99	Passed
Calibration log	True value plus logged noise equals the published value	All seven queries	Passed
Composition	The queries compose to the claimed total	7 × 0.2571 = 1.8	Passed
Reconstruction	Noise has the size the mechanism requires	Ratio 0.94	Passed

The budget audit compared the reported final ε of every private configuration with the last value in its training log. The two values were identical in every case. They were 0.990572, 2.99984 and 7.995129 against targets of 1, 3 and 8. The logged ε never went down from one step to the next. Figure 6.4 compares these values with the targets and the limit.

The reconstruction check compared the published values of the corrected release with the source values. Across the five scalar fields, the mean absolute difference was 4.37. A correct Laplace mechanism at ε = 0.2571 per query gives an expected value of 4.67. The ratio of 0.94 is close to the ideal of 1.0. This shows the noise has the expected size, and the source values cannot be read from the release.

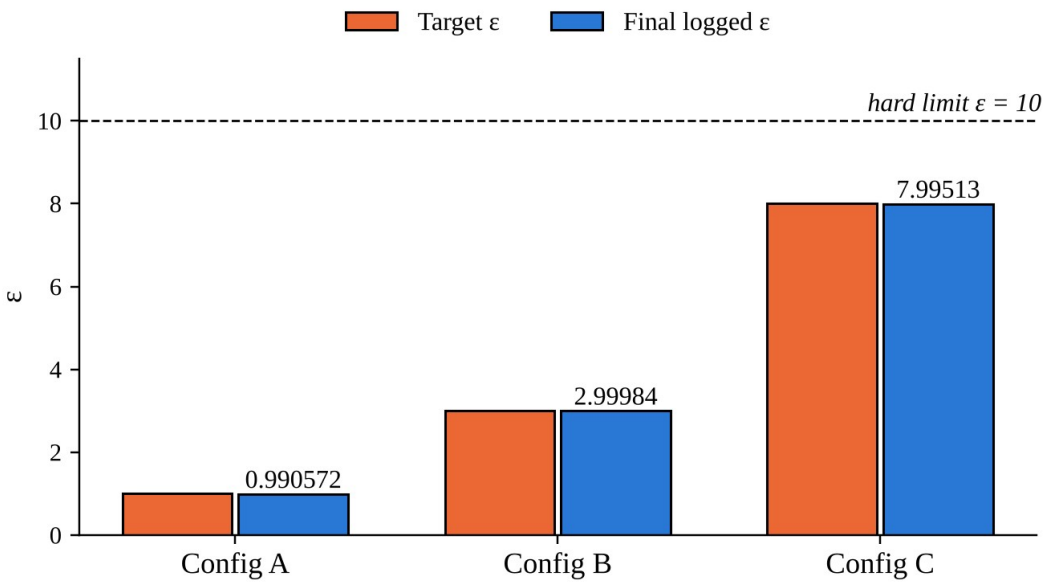


Figure 6.4: Target and final logged ε of the three private configurations (independent audit by Risk and Compliance)

Risk and Compliance verified the rest of the pipeline as well. All 40,000 dataset files matched their SHA-256 manifests. The released metadata had k-anonymity of k = 5 over 10,000 rows. The leakage scan found nothing in 9 metadata files. The compliance certificate R26-RC-F3335EAE8286 gave an overall verdict of PASS. All seven checks passed, and eight of eight GDPR controls were verified. The other components reported the results that the joint certificate needs. Utility Evaluation trained a YOLOv8n detector on the synthetic data. On KITTI, it kept 10.46% of the real-data mAP50, or 6.2% on mAP50-95. The Fréchet Inception Distance (FID) was 115.99. The poisoning gate of Utility Evaluation filtered 24 of 100 label batches. Dataset Generation audited an authenticated AES-256 Galois/Counter Mode (GCM). It rejected a changed file and left none of 6,093 encrypted Town01 images readable without the key.

6.5 Summary of findings

- The Laplace mechanism privatized all seven calibration statistics, and the simulator used only the privatized values. The corrected release stays within a total of ε = 1.8. The dataset was captured from the corrected release.
- Every private training run stayed within its target ε and far below the limit of 10.

- The shadow-model attack succeeded against the baseline (AUC 0.639, success 56.7%). It fell to near chance against every private model (AUC 0.504 to 0.515, success 50.3% to 51.6%). The drop was significant for all three budgets.
- The loss-threshold attack confirmed the result. At a 1% FPR, no model leaked more than chance, including the baseline.
- In the stress test, DP-SGD removed 99.5%, 79.4% and 63.0% of a clear leak at $\varepsilon = 1$, 3 and 8. Every reduction was significant, and smaller budgets gave significantly stronger protection.
- The cost of privacy in the main study was a drop of 0.16 to 0.21 in test AUC. The three budgets did not differ significantly in this cost.
- Risk and Compliance independently verified all twelve reports, the budget logs and the calibration noise. Its composition check found the understated calibration total, which was then corrected.

7 DISCUSSION

7.1 Interpretation of findings

The results support a positive answer to the research question, with some limits. For RQ1, the calibration step privatized all seven statistics within a total of ε = 1.8. The simulator was used in the same way as before. For RQ2, DP-SGD removed the membership signal that the attacks used. The same attack succeeded against the baseline. A broken attack cannot explain the results near 50% on the private models. The stress test showed the same effect against a clear leak. For RQ3, the protection cost a measurable drop of 0.16 to 0.21 in test AUC.

The size of the effect needs careful reading. The drop in attack AUC, from 0.639 to about 0.50, is statistically significant. However, it starts from a baseline that leaked only moderately. A success rate of 56.7% means the attack was right only slightly more often than a coin toss. At a 1% FPR, the loss-threshold attack could not pick out any frame with confidence. This was true even on the baseline. The leakage that DP removed was a small average shift in loss across many images. The review panel raised this point (Section 6.3). The main study alone could not answer it, because no severe leak was present. The stress test addressed this gap. Against a model that memorized its 500 training frames, the attack AUC reached 0.628. DP-SGD at ε = 1 brought it down to 0.501. A weak baseline does not explain the protection.

The role of ε also needs discussion. In the main study, the three budgets gave almost the same result against the attacks. This happened although ε = 8 is a much weaker formal bound than ε = 1. Most of the utility cost came from using DP-SGD at all. The choice of budget added little to it. The stress test explains this result. With little leakage to remove, every budget reached chance. Once the leak was clear, the budgets separated in the order that theory predicts. They removed 99.5%, 79.4% and 63.0% of the advantage at ε = 1, 3 and 8. A looser budget is enough only when the model leaks little.

7.2 Comparison with literature baselines and existing solutions

The gap between the formal bound and the measured leakage matches earlier work. Jayaraman and Evans [10] found that measured leakage at a given ε varies widely. It is often far below the bound. Nasr et al. [11] showed that a worst-case attacker can reach

the bound. Realistic attackers fall well short of it. The attacks in this study are realistic attacks. For this reason, results near chance for ε = 1, 3 and 8 are expected.

The low-FPR result agrees with Carlini et al. [6]. An AUC of 0.639 looks like clear leakage. However, the TPR at a 1% FPR showed that no single record was exposed with confidence. Reporting both measures meant that the baseline leak was neither hidden nor overstated. The overfitting signal also behaved as Yeom et al. [19] described. The baseline had a loss gap of 0.36, and the private models had gaps of 0.001 to 0.026. This matches the fall in attack success.

The component covers gaps left by existing tools. Simulators such as CARLA [4] leave the calibration step unprotected. In this component, the step receives a formal guarantee. Training libraries such as Opacus [20] apply DP-SGD but do not test the trained model. Auditing tools such as ML Privacy Meter [21] measure leakage. They do not apply privacy or link the result to a signed budget record. The component combines these steps into one workflow. Another party can then check its output.

7.3 Research contribution

The main research contribution is a measured link between the formal budget ε and attack resistance. It applies to models trained on synthetic driving images. The study tested three budgets with three seeds each. Each budget faced a positive control and two different attacks. The test was then repeated on an overfitted model with five seeds. All three budgets reduced a moderate leak to chance level. Against a clear leak, the measured protection followed the order of the budgets. These differences were significant.

A second contribution is a method for reporting such results. The method reports the AUC, the success rate and the TPR at low FPR together. It adds the loss gap and paired tests against same-seed baselines. A third contribution is the finding that the calibration channel is a separate privacy point. Seven queries at ε = 1.8 each add up to ε = 12.6 under sequential composition. A per-query budget can look small while the total is large. The corrected module takes the total as its input and splits it over the queries.

7.4 Product contribution

The product contribution is the privacy_integration package. It privatizes calibration statistics, trains models under a hard budget limit, attacks them and signs the results. Its outputs can be checked without repeating any training. During the project, Risk and Compliance used these outputs in its independent check. This check found a reporting error that the author had missed. Without the signed logs and the separate check, this error in the calibration total would have gone unnoticed.

7.5 Limitations and threats to validity

Internal validity. Three seeds per configuration separated the private models from the baseline. They did not separate the three budgets from each other, but the five seeds of the stress test did. The verdict rules of the stress test were changed after its first run. The first version used stricter cut-offs of its own. The final version uses the standard tests in Section 3.5. Under the first rules, the positive control would have failed. The reduction would also not have been shown on every seed. The certificate reports both verdicts for this reason. Opacus secure mode was turned off. The torchcsprng library that it needs is no longer maintained and does not support PyTorch 2.x. As a result, the noise came from a standard random number generator. Positive frames were duplicated to balance the training set. A duplicated frame contributes to more than one gradient, which weakens its per-record guarantee.

Construct validity. Privacy was measured with a binary presence classifier, not with the YOLOv8 detector of Utility Evaluation. The attacks in this study are realistic but not the strongest known. A reference-calibrated attack following Carlini et al. [6] has been implemented. It has not yet been tested, so no result is claimed from it.

External validity. The calibration statistics came from a single source, the Waymo Open Dataset. The results come from one simulator, one model and one task. They may not hold for larger models or other kinds of data.

Utility. Utility Evaluation trained the same detector on synthetic data and on real KITTI images. The synthetic model reached 10.46% of the real-data mAP50, or 6.2% on mAP50-95. The proposal target of staying within 15% of the real-data baseline was not reached. That detector was trained without DP-SGD. The gap does not come from

the training step of this component. The dataset was generated at one calibration budget only. For this reason, the share of the gap caused by calibration noise could not be measured.

7.6 Implications

For practitioners, the results suggest that a formal ε is not enough evidence on its own. A team should attack its own models and include a non-private control. It should report the low-FPR result as well as the AUC. In this study, the small model leaked little, and $\varepsilon = 8$ nearly matched $\varepsilon = 1$. The stress test shows that this advantage disappears when a model memorizes its data. A team should choose the budget for the worst model it expects to train.

For the design of synthetic data pipelines, the calibration channel needs the same care as training. Budgets must be added up over all queries that read the same records. The noise log must also be treated as sensitive. For auditors and regulators, signed reports and per-step budget logs show that privacy evidence can be checked. The checker does not need to trust the team that produced the evidence.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The component gives a privacy record to a team that builds perception models with simulated data. The team can show this record to others. The record turns a formal budget into evidence. It contains a signed budget log and attack results against private and non-private models. It includes a certificate that another party can verify. With this record, a team can use real driving statistics to make its simulation realistic. It can still show that those statistics and the models trained later were protected.

The project proposal identified four groups of users:

- **Autonomous vehicle developers**, mainly machine learning engineers and data infrastructure leads. They need training data at scale without a long privacy review of every dataset.
- **Data protection officers and regulators**, who need evidence they can read and check.
- **Academic researchers**, who want to share synthetic datasets with a clear privacy record.
- **Road users**, whose faces, plates and routes should not be exposed through the training pipeline.

8.2 Market and competitive position

The target market is teams that already use simulation for perception training. Examples are autonomous vehicle start-ups, automotive suppliers, university research labs and simulation service providers. Many of these teams want to use real statistics to make simulations realistic. However, privacy rules and contracts prevent them from sharing raw recordings.

Each competing approach covers only part of the need. Face and plate blurring removes visible identifiers but does not protect the trained model. Simulators such as CARLA and NVIDIA DRIVE Sim produce realistic data. The project proposal found no formal privacy guarantee for their calibration step. Training libraries such as Opacus apply DP-SGD but do not test the result. Auditing tools such as ML Privacy Meter test models but do not apply privacy. This component combines these steps. It

protects the calibration channel and applies DP-SGD. It then attacks the result with a positive control and signs the evidence.

8.3 Business or sustainability model

The most practical model is an open-source core with paid services. The privacy_integration package could be released as an open-source toolkit. This would lower the barrier for research labs and build trust in the method. Companies could pay for integration with their own simulators and training pipelines. They could also pay for support and training. A hosted service could run attacks and produce signed privacy reports for models that customers upload. Free academic use would help the method spread and keep it open to review by others.

The running costs are low. The whole experiment in this thesis ran on a laptop GPU with 4 GB of memory. The main dependencies are free, open-source libraries. The largest cost would be the staff time needed to adapt the component to each customer's data and models.

8.4 Scalability intellectual property and compliance

Scalability. The calibration step is cheap, because it adds noise to a few statistics. The training and attack steps grow with the size of the model and data. Larger detectors would need more GPU memory, because DP-SGD computes a gradient for every sample. Opacus supports larger models. However, each new model would need its own tuning under DP-SGD. This study needed such tuning for the optimizer and the normalization layer.

Intellectual property. The component builds on open-source libraries such as PyTorch, Opacus and CARLA. Any release must respect their licenses and keep their notices. The code written for this component belongs to the author and the project. This is subject to the rules of the university. The calibration statistics came from the Waymo Open Dataset. The group used KITTI only as a utility baseline. Both datasets are licensed for non-commercial use only. A commercial product could not use them under these licenses.

Compliance. The component produces evidence that a data controller can use for compliance. Its evidence maps to GDPR Articles 5(1)(c), 5(2) and 25 [3] and to the NIST Privacy Framework [23]. The final legal decision stays with the data controller. Any product must state its limits clearly. Examples are the missing secure noise source and the duplication of training records. Otherwise, users may trust the guarantee more than they should.

8.5 Contribution to the Sustainable Development Goals

The component contributes most to two Sustainable Development Goals.

- **SDG 9: Industry, Innovation and Infrastructure.** Target 9.5 asks for stronger research and technological capability. The component gives developers a tested way to use synthetic data with a formal privacy record. This supports privacy-preserving work in the automotive sector.
- **SDG 16: Peace, Justice and Strong Institutions.** Target 16.10 includes the protection of fundamental freedoms, and privacy is one of them. The component helps keep personal data out of training pipelines. It also gives regulators evidence they can verify.

The link to road safety under SDG 3 (target 3.6) and SDG 11 (target 11.2) is indirect. Better access to safe training data may help perception models improve. However, this thesis did not measure any effect on safety. There are also possible negative effects. A signed certificate could give a false sense of security. This can happen if users read ε without the attack results and limitations. Training many models uses energy as well. Publishing the attack results and limitations with every certificate reduces the first risk. Keeping models small where possible reduces the second.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This research made the privacy of synthetic autonomous vehicle data measurable and open to checks by others. The Privacy Integration component applies the Laplace mechanism to the statistics that configure CARLA. It applies Opacus DP-SGD with an RDP accountant to the models trained on the generated frames. It then attacks its own models and signs the results.

Over three seeds, the shadow-model attack reached a success rate of 56.7% against the unprotected model. Against every private model, it reached only 50.3% to 51.6%. The cost was 0.16 to 0.21 in test AUC. A second loss-threshold attack gave the same result. Neither attack could identify individual frames with high confidence for any model. In short, DP-SGD reduced a moderate leak to chance level. The three budgets gave almost the same protection. A stress test answered the question that this result left open. The test used a model that memorized its training frames, with five seeds. DP-SGD removed 99.5%, 79.4% and 63.0% of the attacker's advantage at $\varepsilon = 1$, 3 and 8. The budgets separated in the order that theory predicts. The signed reports also allowed Risk and Compliance to check the evidence on its own. This check found an error in the reported calibration total. The corrected module now keeps all seven queries within a total of $\varepsilon = 1.8$.

9.2 Achievement of objectives

Table 9.1 compares the results with the objectives in Section 1.5.

Table 9.1: Achievement of objectives

Objective	Outcome	Status
Select a DP training library	Opacus 1.4.1 with per-sample gradients and RDP accounting	Achieved
Calibration budget below $\varepsilon = 2$	Corrected release composes to 1.8, and the dataset was captured from it	Achieved
DP-SGD with an automatic limit	All runs within target ε (0.991 to 7.995) and the limit of 10	Achieved
Attack success at or below 55%	50.3% to 51.6% with DP, 56.7% without	Achieved

Objective	Outcome	Status
Privacy and utility over budgets and seeds	Three budgets with three seeds each, and a five-seed stress test	Achieved
Signed evidence verified by another party	12 of 12 reports, the budget logs and the calibration noise verified by Risk and Compliance	Achieved
Utility within 15% of real data (group target)	10.46% of the real-data mAP50 (Utility Evaluation)	Not reached

9.3 Individual research and product contributions

The individual research contributions of this thesis are as follows:

- A measured test of how the formal budget ε relates to membership inference on synthetic driving images. The test used a positive control and two attacks.
- Evidence that ε = 1, 3 and 8 all reduce a moderate leak to chance level at a similar utility cost. Against a clear leak, the protection follows the order of the budgets.
- A reporting method that combines AUC, success rate, TPR at low FPR, loss gap and paired tests.
- Evidence that the budgets of all calibration queries must be added up. Seven queries at ε = 1.8 each give a total of 12.6. For this reason, the corrected module splits a total budget over the queries.

The individual product contributions are as follows:

- The privacy_integration package with its six modules.
- The DP training wrapper with a hard budget limit.
- The classifier and label rule that made private training work on the CARLA frames.
- The stress-test and reference-calibrated attack commands.
- The signed audit reports and the privacy certificate in JSON and PDF form.
- The JSON contracts that connect the component to the rest of the framework, and 135 unit tests.

9.4 Future directions

The future work is listed in order of priority.

1. **Stronger attacks.** Run the implemented reference-calibrated attack with more reference models, and report the TPR at 0.1% FPR.
2. **More seeds.** Run more seeds per configuration, so that the main study can also separate the three budgets.
3. **Training on filtered labels.** Use the existing --labels-dir hook to train on the labels that passed the poisoning gate of Utility Evaluation. Then measure the effect on utility.
4. **Secure noise and less duplication.** Use a cryptographically secure random number generator that works with current PyTorch versions. Balance the training set without duplicating records.
5. **Public-key signatures.** Replace HMAC with public-key signatures. Any third party could then verify the reports without a shared key.

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GLOSSARY

Differential privacy. A property of a randomized computation that limits how much any single record can change its output.

Privacy budget (ε). The number that sets the strength of differential privacy. A smaller ε means stronger protection and more noise.

Failure probability (δ). The small probability with which the ε limit of (ε, δ)-differential privacy may fail.

Sensitivity (Δf). The largest change that one record can cause in the answer to a query.

Sequential composition. The rule that the budgets of several private queries on the same records add up.

Post-processing. Any further computation on a private output, which cannot weaken its guarantee.

Noise multiplier (σ). The ratio of the standard deviation of the DP-SGD noise to the clipping norm C.

Privacy accountant. A component that tracks the cumulative ε spent during training.

Membership inference. An attack that tries to tell whether a given record was in the training set of a model.

Shadow model. A model trained by an attacker to copy the behavior of a target model.

Positive control. A case in which the attack is expected to succeed, used to show that the attack works.

Loss gap. The difference between the mean loss of unseen records and that of training records.

APPENDICES

Appendix A Extended results

This appendix gives the signed evidence and the calibration audit in more detail than Chapter 6. Table A.1 lists the privacy record of each private configuration. The values come from the signed audit reports and the independent audit.

Table A.1: Privacy record of each private configuration (seed 42)

Run	Target ε	σ	Final logged ε	Steps logged	Attack AUC in report
Config A	1.0	0.977	0.990572	3,090	0.5044
Config B	3.0	0.665	2.99984	3,090	0.5144
Config C	8.0	0.507	7.995129	3,090	0.5131

Each final logged ε is slightly below its target. Opacus chooses σ so that the budget at the final step does not pass the target. In the tamper test, the recorded ε in a copy of the Config A report was changed from 0.990572 to 0.01. Verification failed at once.

Table A.2 summarizes the reconstruction check on the corrected calibration release. Risk and Compliance carried out this check with the private audit log.

Table A.2: Reconstruction check on the calibration channel (corrected release)

Measure	Value
Mean absolute difference between published and source values (five scalar fields)	4.37
Expected mean absolute difference for a correct Laplace mechanism	4.67
Ratio of observed to expected difference	0.94
Largest relative distortion (intersections per km)	537.8%
ε per query and total under sequential composition	0.2571 and 1.8

Table A.3 lists the stress-test result of every seed.

Table A.3: Stress-test attack AUC per seed (95% bootstrap interval)

Seed	No DP	ε = 1	ε = 3	ε = 8
42	0.630 (0.595 to 0.665)	0.485 (0.447 to 0.522)	0.516 (0.480 to 0.552)	0.542 (0.506 to 0.578)
43	0.636 (0.602 to 0.671)	0.495 (0.460 to 0.533)	0.548 (0.512 to 0.583)	0.574 (0.539 to 0.608)
44	0.605 (0.566 to 0.637)	0.499 (0.465 to 0.532)	0.518 (0.481 to 0.552)	0.541 (0.508 to 0.574)
45	0.612 (0.578 to 0.646)	0.501 (0.467 to 0.533)	0.513 (0.475 to 0.548)	0.543 (0.506 to 0.577)
46	0.658 (0.622 to 0.691)	0.524 (0.490 to 0.560)	0.537 (0.501 to 0.574)	0.536 (0.500 to 0.573)

Appendix B Academic integrity and AI use disclosure

Generative AI was used while preparing this thesis, in line with the module policy on AI use.

Tool. Claude, a large language model by Anthropic.

Use. The tool drew the diagrams and charts from the author's results. It was also used for a final verification of the thesis before submission.

Verification. The author checked the numbers, citations and claims against the experiment outputs and the cited sources.

Responsibility. The research design, the experiments, the results and the conclusions are the responsibility of the author.